



## Handout 6.7: Quick Bread Lab Sheet

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

Station: \_\_\_\_\_ My role: \_\_\_\_\_

Print for every student. Bring it both days. Day 1 is pages 1 to 3, Day 2 is pages 4 and 5.

## DAY 1

### Part 1. How leavening works

Write it in four sentences, in your own words.







**Which one needs an acid to work? Circle: baking powder / baking soda**

**Where does the acid come from in a real recipe? Name three sources:**

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_

**Part 2. The four-cup fizz test**

| Cup | What is in it           | What happened (none, a little, a lot) |
|-----|-------------------------|---------------------------------------|
| 1   | Baking soda + water     |                                       |
| 2   | Baking soda + vinegar   |                                       |
| 3   | Baking powder + water   |                                       |
| 4   | Baking powder + vinegar |                                       |

**The rule, in one sentence:**

Baking soda needs \_\_\_\_\_, and baking powder does not because

## Part 3. The experiment setup

### The four batches

| Batch       | Leavening treatment        | What I predict |
|-------------|----------------------------|----------------|
| A (control) | 1 tsp baking powder        |                |
| B           | none at all                |                |
| C           | 2 tsp baking powder        |                |
| D           | 1 tsp baking soda, no acid |                |

### Variables

**Independent variable** (the one thing we change on purpose):

**Dependent variable** (what we measure to see the result). Name all three:

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_

**Constants** (everything we keep exactly the same). List at least four, and go for eight:

|    |    |
|----|----|
| 1. | 5. |
| 2. | 6. |
| 3. | 7. |
| 4. | 8. |

**Why do we need Batch A at all, if we already know a normal muffin rises?**



#### Part 4. My hypothesis and prediction

**My hypothesis** (one sentence, in the form “if we change , *then* ”):





**My prediction: rank the four batches by how high I think they will rise.**

Highest \_\_\_\_ second \_\_\_\_ third \_\_\_\_ lowest \_\_\_\_

**Batch D prediction:** what will it taste like and why?



## Day 1 exit card

The one thing we are changing is \_\_\_\_\_.

Batch \_\_\_\_\_ will rise the highest because \_\_\_\_\_.

## DAY 2

### Part 5. Data

Measure the batter height in the liner before baking and the muffin height after. Use the ruler.

| Batch                | Height<br>BEFORE<br>baking | Height<br>AFTER<br>baking | How much it<br>rose | Crumb<br>(none, few,<br>or many<br>holes) | Taste |
|----------------------|----------------------------|---------------------------|---------------------|-------------------------------------------|-------|
| A (control)          |                            |                           |                     |                                           |       |
| B (no<br>leavening)  |                            |                           |                     |                                           |       |
| C (double)           |                            |                           |                     |                                           |       |
| D (soda, no<br>acid) |                            |                           |                     |                                           |       |

**Time the tin went in: \_\_\_\_\_ Time it came out: \_\_\_\_\_ Oven temperature: \_\_\_\_\_**

#### Actual ranking by height

Highest \_\_\_\_\_ second \_\_\_\_\_ third \_\_\_\_\_ lowest \_\_\_\_\_

#### Compare to your prediction

| Batch | I predicted | What actually<br>happened | Same or different? |
|-------|-------------|---------------------------|--------------------|
| A     |             |                           |                    |
| B     |             |                           |                    |
| C     |             |                           |                    |
| D     |             |                           |                    |

### Part 6. Conclusion

Fill in all four parts.

**The data showed** \_\_\_\_\_



**My hypothesis was** (circle one) **supported** / **not supported** / **partly supported**, because



**My evidence is** \_\_\_\_\_



**One thing that could have gone wrong with our test is \_\_\_\_\_**



**One sentence our station can read out loud at the station check:**



### Station check

| Line                                                          | Done? |
|---------------------------------------------------------------|-------|
| Oven off, tin on the trivet, nothing hot left out             |       |
| All four bowls washed, dried, and put away per the drawer map |       |
| Measuring cups and spoons washed and put away                 |       |
| Counters wiped and sanitized                                  |       |
| Sink empty and rinsed                                         |       |
| Floor swept, trash out                                        |       |
| Leavening jars returned to the teacher's table                |       |

Teacher signature: \_\_\_\_\_



## Grade 8 stretch: Batch E

**Batch E: 1 tsp baking soda PLUS 1 tsp white vinegar.** Everything else the same.

| Batch E | Before | After | How much it<br>rose | Crumb | Taste |
|---------|--------|-------|---------------------|-------|-------|
|         |        |       |                     |       |       |

**Does Batch E behave more like Batch A or more like Batch D? \_\_\_\_\_**

**What does that prove? \_\_\_\_\_**



**Name one confounding variable in our own test** (something we did not control that could have affected the result), **and say how you would control it next time.**

The variable: \_\_\_\_\_

How to control it: \_\_\_\_\_